PixCuboid: Room Layout Estimation from Multi-view Featuremetric Alignment

GUSTAV HANNING, KALLE ÅSTRÖM, VIKTOR LARSSON

LUND UNIVERSITY

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Summary

- We propose a novel multi-view room layout estimation method based on featuremetric alignment.
  - **Input**: A collection of posed images.
  - **Output**: The predicted room layout (a cuboid).
- Using multiple views resolves the global scale.
- With our flexible, optimization-based approach we can:
  - Easily change the cost function.
  - Do multi-room layout estimation.
  - Refine other geometries, not just cuboids.

Datasets

We create two datasets, based on ScanNet++ v2 and 2D-3D-Semantics, to benchmark multi-view cuboid room layout estimation.

- **Ground truth layouts**: For each scene, a cuboid is fitted to vertices labeled "floor", "wall" and "ceiling" in the semantic mesh. We manually inspect the results and discard non-cuboid scenes.
- **Images**: For ScanNet++ we sample image tuples, consisting of ten random DSLR images, to use as input. In 2D-3D-Semantics we randomly pick two panoramas, each split into four perspective images, for every room.

Dataset	Train	Validation	Test
ScanNet++ v2	391	10	18
2D-3D-Semantics	0	0	160

Table 1. Number of scenes for the two datasets.

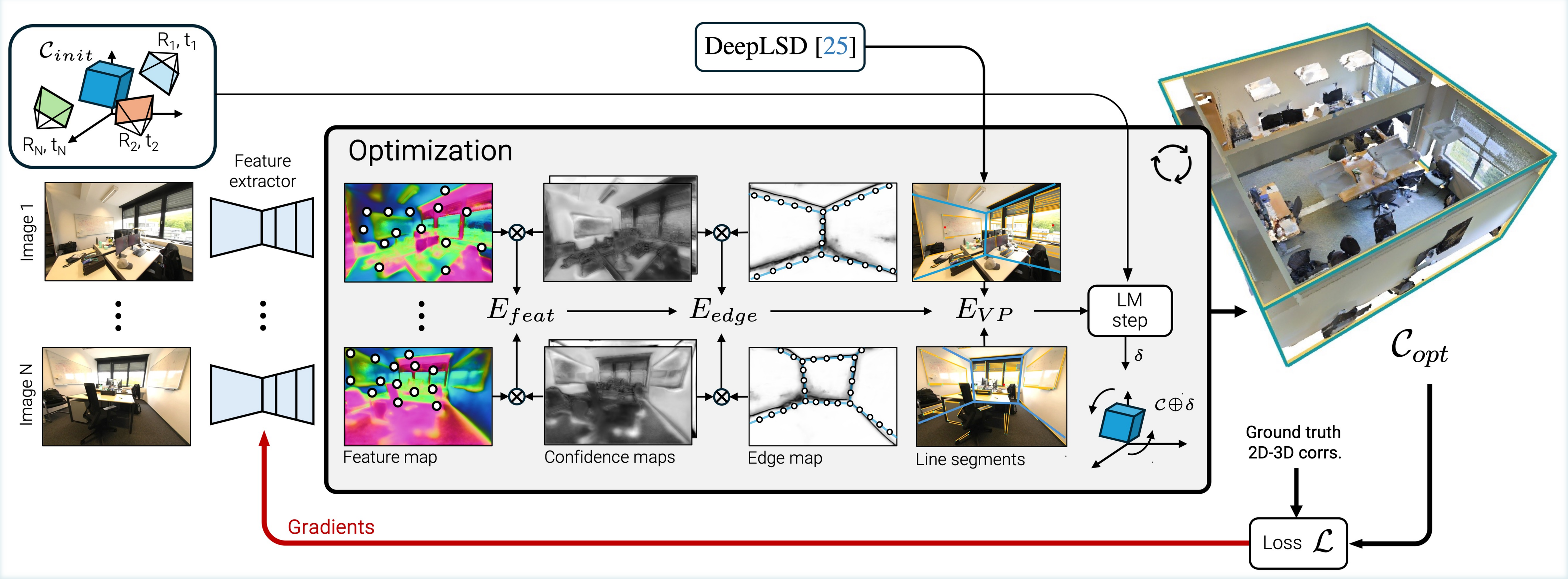

Figure 1.Taking only posed images as input, PixCuboid estimates the room layout $\mathcal{C}_{\mathrm{opt}}$ by optimizing from a coarse initial estimate $\mathcal{C}_{\mathrm{init}}$. First, a deep network predicts dense feature-, confidence- and edge maps for each image. The optimization then minimizes a combination of three terms (see below). The network is trained end-to-end by supervising on the result of the optimization, propagating gradients back to the network through the optimization steps.

Cuboid Optimization

To refine the cuboid $\mathcal{C}$ we define a cost function $\mathrm{E}(\mathcal{C})$ depending on three terms:

a) A featuremetric cost $\mathrm{E}_{\mathrm{feat}}(\mathcal{C}) = \sum_{\mathrm{i,j}} \sum_{\mathrm{k}} \mathrm{w}_{\mathrm{ijk}} \rho \left(\| \mathbf{F}_{\mathrm{i}}[\mathbf{x}_{\mathrm{ik}}] - \mathbf{F}_{\mathrm{j}}[\mathcal{W}_{\mathrm{i}\to\mathrm{j}}(\mathbf{x}_{\mathrm{ik}}, \mathcal{C})] \|^2 \right)$.

b) A monocular edge cost $\mathrm{E}_{\mathrm{edge}}(\mathcal{C}) = \sum_{\mathrm{i}} \sum_{\mathrm{j}} \mathrm{w}_{\mathrm{ij}} \mathbf{E}_{\mathrm{i}}\left[\Pi_{\mathrm{i}}\left(\mathbf{R}_{\mathrm{i}}\mathbf{X}_{\mathrm{j}} + \mathbf{t}_{\mathrm{i}}\right)\right]^2$.

c) A VP-based cost $\mathrm{E}_{\mathrm{VP}}(\mathcal{C}) = \sum_{\mathrm{i}} \sum_{\mathrm{j}} \min\left(\min_{\mathrm{k}\in\{1,2,3\}} \mathrm{D}_{\mathrm{VP}}\left(\mathbf{l}_{\mathrm{ij}}, \mathbf{v}_{\mathrm{ik}}\right), \tau \right)^2$.

We minimize $\mathrm{E}(\mathcal{C}) = \mathrm{E}_{\mathrm{feat}}(\mathcal{C}) + \alpha \mathrm{E}_{\mathrm{edge}}(\mathcal{C}) + \beta \mathrm{E}_{\mathrm{VP}}(\mathcal{C})$ using standard Levenberg-Marquardt optimization.

- **Multi-scale**: The optimization is performed in a coarse-to-fine manner, with increasingly higher resolution feature maps $\mathbf{F}_{\mathrm{i}}$ and edge maps $\mathbf{E}_{\mathrm{i}}$.
- **Initialization**: We use a simple heuristic to initialize $\mathcal{C}$ from the known camera poses.

Training

PixCuboid is trained on our ScanNet++ v2 training set with the loss

$$\mathcal{L}(\mathcal{C}) = \frac{1}{\mathrm{N}} \sum_{\mathrm{i,j}} \sum_{\mathrm{k}} \| \mathcal{W}_{\mathrm{i}\to\mathrm{j}}\left(\mathbf{x}_{\mathrm{ik}}^{\mathrm{GT}}, \mathcal{C}\right) - \Pi_{\mathrm{j}}\left(\mathbf{R}_{\mathrm{j}}\mathbf{X}_{\mathrm{ik}}^{\mathrm{GT}} + \mathbf{t}_{\mathrm{j}}\right) \|^2.$$

- $\{(\mathbf{x}_{\mathrm{ik}}^{\mathrm{GT}}, \mathbf{X}_{\mathrm{ik}}^{\mathrm{GT}})\}$: 2D-3D correspondences on the floor/wall/ceiling.

Results

We evaluate on 2D-3D-Semantics, comparing against both single- and multi-view methods (none of which are trained on this dataset).

Method	Input	IoU ↑	Chamfer ↓	Rotation ↓	Depth ↓	Normal ↑
Total3D	1 x	31.8	1.57 m	8.9°	1.41 m	44.1
Implicit3D	1 x	31.6	1.59 m	10.4°	1.51 m	37.6
Deep3DLayout	1 x	58.8	0.68 m	N/A	0.44 m	52.6
LED²-Net	1 x	68.7	0.45 m	N/A	0.40 m	34.6
PSMNet	2 x	43.5	1.04 m	N/A	0.63 m	51.4
PixCuboid (ours)	8 x	**89.0**	**0.18 m**	**0.7°**	**0.10 m**	**96.1**

Table 2. Room layout estimation results on the cuboid-shaped spaces of 2D-3D-S.

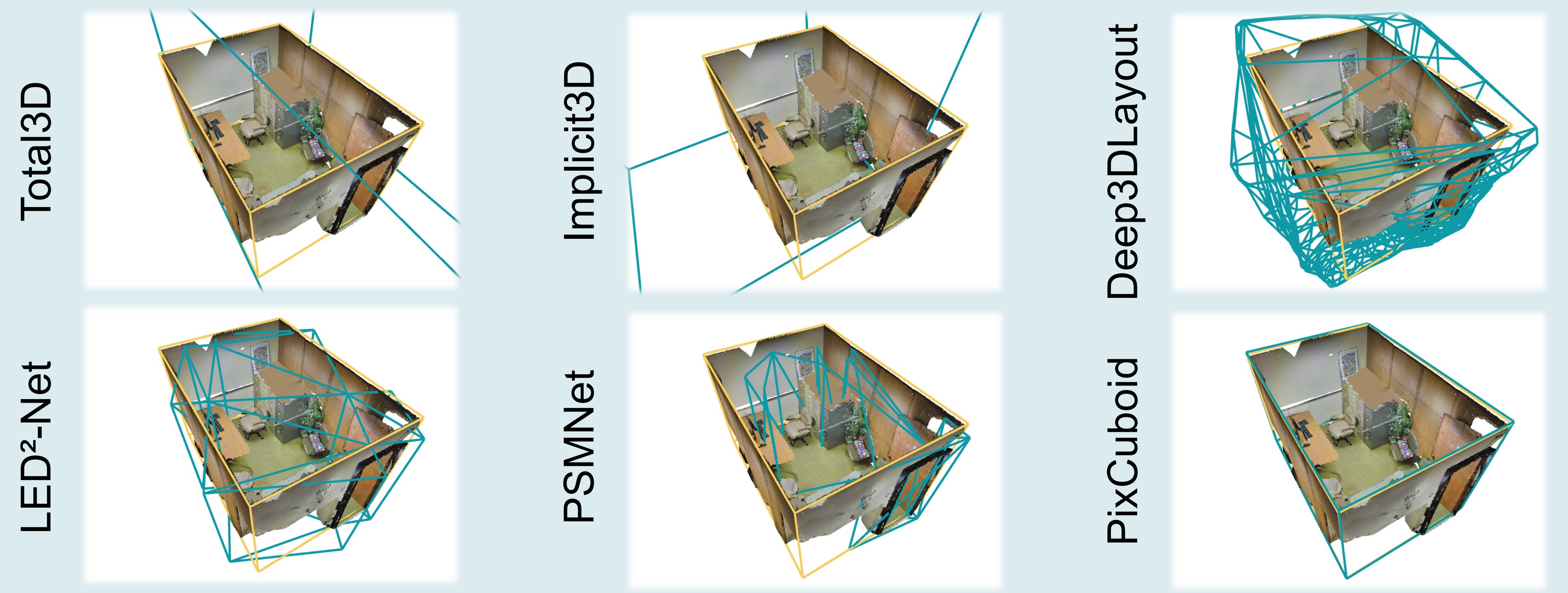

Figure 2. Qualitative comparisons of predicted room layouts for one space in 2D-3D-Semantics. Predictions are shown in blue and the ground truth cuboids in yellow.